

EXPERIMENT-1

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FRAME 1:

INSTRUCTION PAGE:

Instruction Page - Chunking Analysis

Chunking is a cognitive strategy that breaks information into smaller, meaningful units, improving comprehension and memory retention. The Instruction Page of the Visual Memory Recall UI effectively applies chunking through the following design choices:

1. Clear and Sequential Information Structure

- Instructions are presented in short, clearly separated sections rather than long paragraphs.
 - Each instruction communicates one primary idea, helping users process information step-by-step.
 - The flow prepares users mentally for the upcoming task without overwhelming them.
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2. Logical Grouping of Concepts

- **Task Overview** - Explains the purpose of visual learning and memory recall.
- **Learning Approach** - Introduces the idea of grouped concepts (chunking).
- **Action Prompt** - Encourages users to begin the task using a clear call-to-action.

This grouping ensures users understand *what* they are doing, *why* they are doing it, and *what comes next*.

3. Visual Hierarchy and UI Elements

- Large, bold title: "**Visual Learning Boost**" clearly communicates the purpose.
- Subtitle provides concise supporting context.
- Friendly brain illustration reinforces the learning theme.

- Prominent **"Get Started"** button clearly signals progression.
- Supporting icons (Organize, Understand, Recognize, Review) visually reinforce learning stages.

4. Simplicity and Clarity

- Minimal text avoids cognitive overload.
- High contrast and spacing improve readability.
- Soft background colors keep attention focused on key content.

Impact: The instruction page primes users for effective chunk-based learning while maintaining clarity and engagement.



FRAME 2:

CHUNKING PHASE:

Chunking Phase - Visual Encoding Screen

This screen represents the **visual encoding stage**, where users observe grouped information and mentally organize items before recall.

1. Purpose of the Screen

- Allows users to visually scan and memorize grouped icons.
 - Encourages pattern recognition and mental categorization.
 - Introduces chunking by grouping related items together.
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2. Key UI Components

Categorized Groups

- Icons are grouped under clearly labeled categories such as:
 - **Social Media**
 - **Study Hub**
 - **Others**
- Each category uses color-coded containers to visually separate chunks.

Icon Grid

- Icons within each group share contextual relevance.
- Familiar visual symbols allow faster recognition.
- Grouping reduces the number of independent elements users must remember.

Action Button

- **"Ready for Review"** clearly signals the transition from encoding to recall.
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3. How Chunking Works Here

1. Users scan icons category by category instead of individually.
2. Items are mentally grouped based on meaning and context.
3. Chunking reduces memory load and improves encoding efficiency.

4. Cognitive G UX Benefits

- Improves short-term memory retention.
- Reduces visual clutter through structured grouping.



FRAME 3:

RECALL PHASE:

Recall Phase - Memory Selection Screen

This screen evaluates how effectively chunking supported memory retention.

1. Purpose of the Screen

- Tests user recall accuracy.
 - Measures the effectiveness of visual chunking strategies.
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2. Key UI Components

Title and Instructions

- Clear heading: **"Check Your Memory"**
- Instructional text guides users to select remembered items.

Icon Selection Grid

- Displays a mix of:
 - Previously seen icons
 - Distractor icons (new items)
- Each icon includes a checkbox for selection.

Submit Button

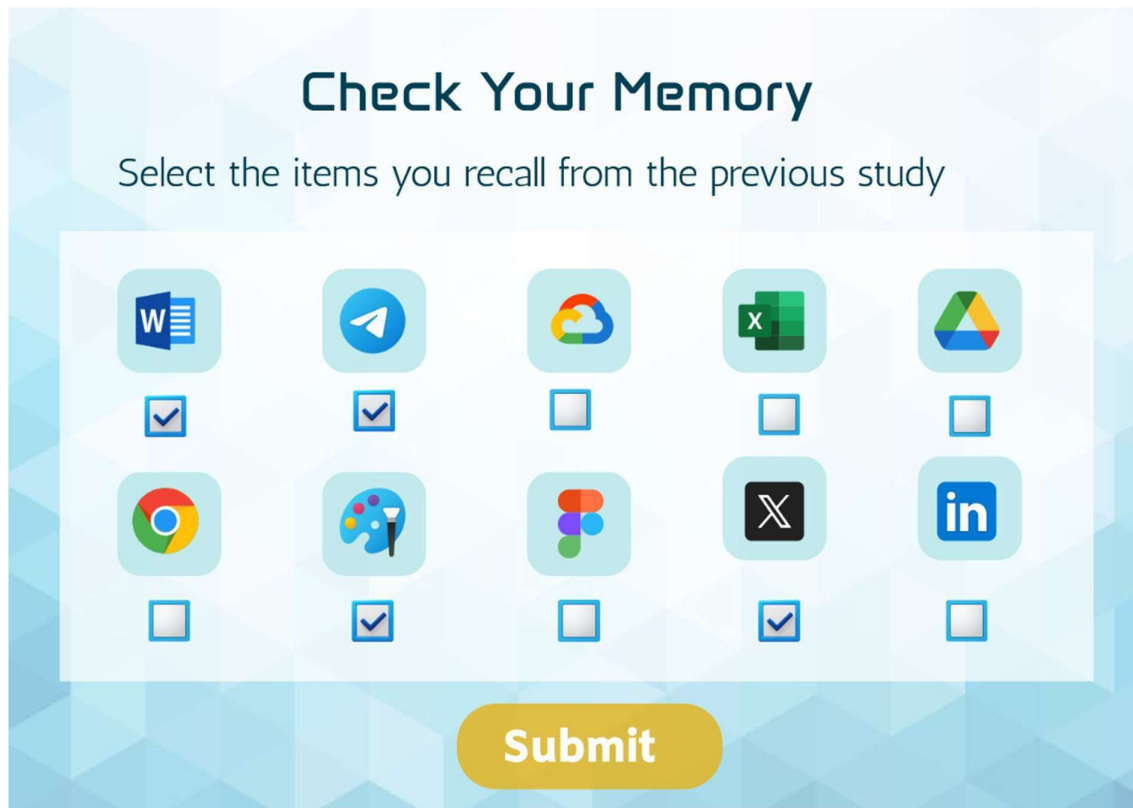
- **"Submit"** button finalizes user choices.
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3. How the Recall Phase Works

1. Users recall categories and icons from the Chunking Phase.
 2. Chunk-based memory allows users to recall groups instead of individual items.
 3. Distractors test false memory and attention accuracy.
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4. Cognitive G UX Benefits

- Tests differentiation between learned and new items.
- Clear interaction design minimizes confusion during recall.



FRAME 4:

RESULT PAGE:

Result Page - Performance Feedback Screen

This screen provides immediate feedback on recall accuracy.

1. Purpose of the Screen

- Displays user performance.
- Reinforces learning through feedback.
- Allows users to continue, review, or exit.

2. Key UI Components

Score Visualization

- Circular progress indicator displaying percentage score (e.g., **80%**).

- Text feedback such as "**Great Job!**" provides positive reinforcement.
- Additional insight shows how many items were correctly recalled.

Feedback Details

- Highlights missed items (optional review opportunity).
- Encourages reflection rather than punishment.

Action Buttons

- **Review** - Allows users to revisit missed content.
- **Exit** - Ends the session.

3. Cognitive G UX Benefits

- Immediate feedback strengthens learning retention.
- Encourages iterative improvement.
- Clear next-step options enhance user control.



PROTOTYPE LINK:

<https://www.figma.com/proto/lt2HED1XEa2ddExzbWa0lj/website?page-id=0%3A1Cnode-id=1-2Cviewport=336%2C143%2C0.18Ct=xKPTiHrNWmAd7mrq-1Cscaling=containCcontent-scaling=fixedCstarting-point-node-id=1%3A2>